IMPERIAL

Department of Mechanical Engineering

Advanced Mechanical Engineering Course

Analytical and Computational Investigation of Noise Radiation and Propagation from Lifting Devices

By

Muhammad Abi Rizky

Bachelor of Engineering, Mechanical Engineering, Universitas Indonesia

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Abstract

Open rotor aeroengines offer significant improvements in propulsion efficiency, but expose fan-generated noise directly to the atmosphere. Predicting this noise requires robust acoustic solvers. This report outlines the preliminary development and validation of an analytical Ffowcs Williams-Hawkings (FW-H) code using a permeable surface, retarded-time formulation. The numerical integration logic was evaluated against exact analytical derivations for both monopole and dipole sources under stationary, co-translating, and relative-motion conditions.

The code demonstrated good accuracy ($< 1\%$ error) for monopole sources, in which it captures convective terms and Doppler phase shifting, given sufficient grid and temporal resolutions. For dipole sources, derived analytical solutions were mathematically proven to fail when physical assumptions are violated. To bypass these limitations, a dual-monopole approximation was implemented and validated. This approach is sufficiently robust across all regimes, as it preserves the convective amplification factors that explicit Taylor expansions omit that was required when defining the dipole source term. While some validation work is still necessary, future work will move to a hybrid computational-analytical framework, using Unsteady Reynolds Averaged Navier Stokes (URANS)-FW-H hybrid to predict open rotor noise generation and propagation.

# Acknowledgements

LOREM IPSUM

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Nomenclature

Variable	**Definition**
c	speed of sound
M	Mach number ($M = u_i/c$)
p	pressure
ρ	density
u	velocity
t	observer time
τ	retarded time
$\Box^2$	d'Alembertian wave operator, $\Box^2 = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$
∇^2	Laplacian differential operator, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
δ	Dirac delta function. Subchapter 2.5: infinitesimally small distance between two monopoles
δ_{ij}	Kronecker delta, $i = j$, $\delta_{ij} = 1$; $i \neq j$, $\delta_{ij} = 0$
y	static coordinate system
η	moving coordinate system
J	Jacobian of $y \to \eta$ coordinate change
$q(x, t)$	monopole source strength
$f_i(x, t)$	dipole source strength. Appendix C, arbitrary response.
T_{ij}	Lighthill stress Tensor, $T_{ij} = \rho u_i u_j + p_{ij} - c^2 p' \delta_{ij}$
λ	wavelength
f	wave frequency
$\hat{r}$	control surface radius
n	normal direction vector
ψ	Appendix C, arbitrary wave
m	Tyler-Sofrin spatial mode
n	Tyler-Sofrin temporal mode
θ	Azimuthal angle
ϕ	Phase angle
ω	Interaction frequency or source pulsation frequency, context are provided each time
Ω	Shaft speed

Variable	Definition (continued)
B	Rotor blade count
V	Stator blade count
Subscripts	
0	value in stationary flow
r	resolved in the direction of radiation
min	minimum/shortest value of interest
max	maximum/largest value of interest
$mono$	monopole source
$dipole$	dipole source
ref	reference value
num	numerical value
i	vector towards the i-direction
j	vector towards the j-direction. Subchapter 2.2 the j-th index of a sum
m	m-th Tyler-Sofrin spatial mode
n	n-th Tyler-Sofrin temporal mode
$fore$	Front blade row
aft	Aft blade row
Superscripts	
$'$	perturbation component, e.g., $p' = p(x,t) - p_\infty$

Chapter 1

Introduction

1.1 Motivation

Due to environmental concerns and the availability of fossil fuels, the efficiency of civil aeroengines have continuously been improved over the last few decades. To improve efficiency, aeroengines have adopted high bypass ratio (BPR) [1], which is supported by the increasing nacelle for bypass air and larger fan diameter. The open rotor concept takes this further by putting the fan outside of the nacelle, which may reduce fuel consumptions and carbon emissions by 10-20% [2] while producing more thrust than existing engines.

However, aeroengine nacelles are designed to suppress the fan-generated noise through the use of acoustic linings, so this concept exposes it directly into the atmosphere, causing high noise pollution. An investigation of this noise generation and its propagation is required to develop suppression methods to reduce this noise. In this work, a hybrid approach between the Ffowcs Williams-Hawkings (FW-H) method and CFD will be used to predict the noise generation and propagation of an open rotor engine.

1.2 Open Rotor Noise Background

Unlike conventional turbofan engines, open rotor architectures lack a nacelle with acoustic liners to suppress the noise. Therefore, the noise generated by the rotors is radiated directly into the atmosphere without suppression. Open rotor noise is caused by the aerodynamic interactions and acoustic interference (inter-rotor and inter-mode interferences) [3]. The reason for this is the acoustic pressure fields generated by rotating blade rows and their aerodynamic interactions. to understand the modal structure of these interactions, the Tyler-Sofrin theory for rotor-stator wake interaction is used as the foundation [4].

The acoustic effect of a single blade passing a stationary vane can be described as a periodic pressure fluctuation repeating at the blade-passage frequency. At a reference position, this fluctuation is expressed as a Fourier series in time, which can then be expanded into a double Fourier series in space and time to describe the spatial-temporal components of the pressure field. For a single interaction, this component is defined as,

$$p'_{m,n} = a_{m,n}\cos(m\theta - nB\Omega t + \phi_{m,n}) \tag{1.1}$$

which represents a traveling wave with m lobes.

When this single-event model is superposed in an array of V evenly spaced stator vanes, the pressure contributions from each vane are shifted in space by $\Delta\theta = 2\pi/V$ and in time by $\Delta t = \Delta\theta/\Omega$. Most of these interferences are destructive; however, when spatial and temporal shifts are aligned, the lobe number m is conserved to satisfy the Tyler-Sofrin mode,

$$m = nB + kV \tag{1.2}$$

where k is any positive or negative integer.

Extending this to the contra-rotating open rotor architecture, this mechanism governs the inter-rotor interaction, with the kinematics being driven by two independently rotating blade rows. By using the aft blade as the reference frame, it acts as a stationary stator where $V = B_{aft}$ vanes spinning at an absolute angular velocity of Ω_{aft}. The front rotor, having B_1 blades, interacts with this aft row at a relative angular velocity of $\Omega_{fore} - \Omega_{aft}$. Applying the Tyler-Sofrin condition in this relative frame changes Equation 1.2 into $m = nB_{fore} + kB_{aft}$. Transforming this back to the absolute stationary frame shows the interaction frequency by adding the reference frame's rotation,

$$\omega = nB_{fore}\Omega_{fore} + kB_{aft}\Omega_{aft} \tag{1.3}$$

This kinematic relationship is why open rotors generate such loud sound signatures. The paper by Dunn, et al., [5] examines the noise data indicating that the frequencies corresponding to shaft orders dominate the sound field. Because the propagation of this noise is governed by the blade counts and rotational speeds, the sound field is sensitive to the engine's kinematic design. A design of experiment study using CFD was conducted to compare the proportionality of various design parameters, such as rotor tip speeds, speed ratio, number of blades, etc. [6]. This is key in understanding that the aeroacoustics analysis can be done through computational methods to design open rotors.

1.3 Hybrid Noise Prediction Background

Predicting open rotor noise requires solving the inhomogeneous wave equation. The FW-H equation provides the standard analytical framework for this process [7]. In this work, the permeable surface formulation (Formulation 1A) is utilised [8, 9, 10]. This formulation was chosen as it captures all noise sources, including quadrupoles (sounds from non-linearities such as shocks and turbulence), via surface integration, provided the control surface fully encapsulates the non-linearities [11], which avoids the computationally expensive volume integrals prohibitive to noise prediction.

In modelling acoustic wave propagation, a few methods are available such as Direct Noise Computation (DNC), FW-H analogy, and Hanson's model, to name a few. It is most practical, however, to conduct the analysis through a hybrid approach, in which flow fields surrounding the noise source are computed using CFD, and the far-field acoustic propagation are computed separately, such as using the FW-H method [12, 13]. This is to avoid the high computational cost of directly resolving the acoustic waves, while having a good geometric sensitivity of the sound source [14, 15, 16], unlike pure analytical approach. The schematic below in Figure 1.1 shows the available methods as well as highlights the current method used in this work.

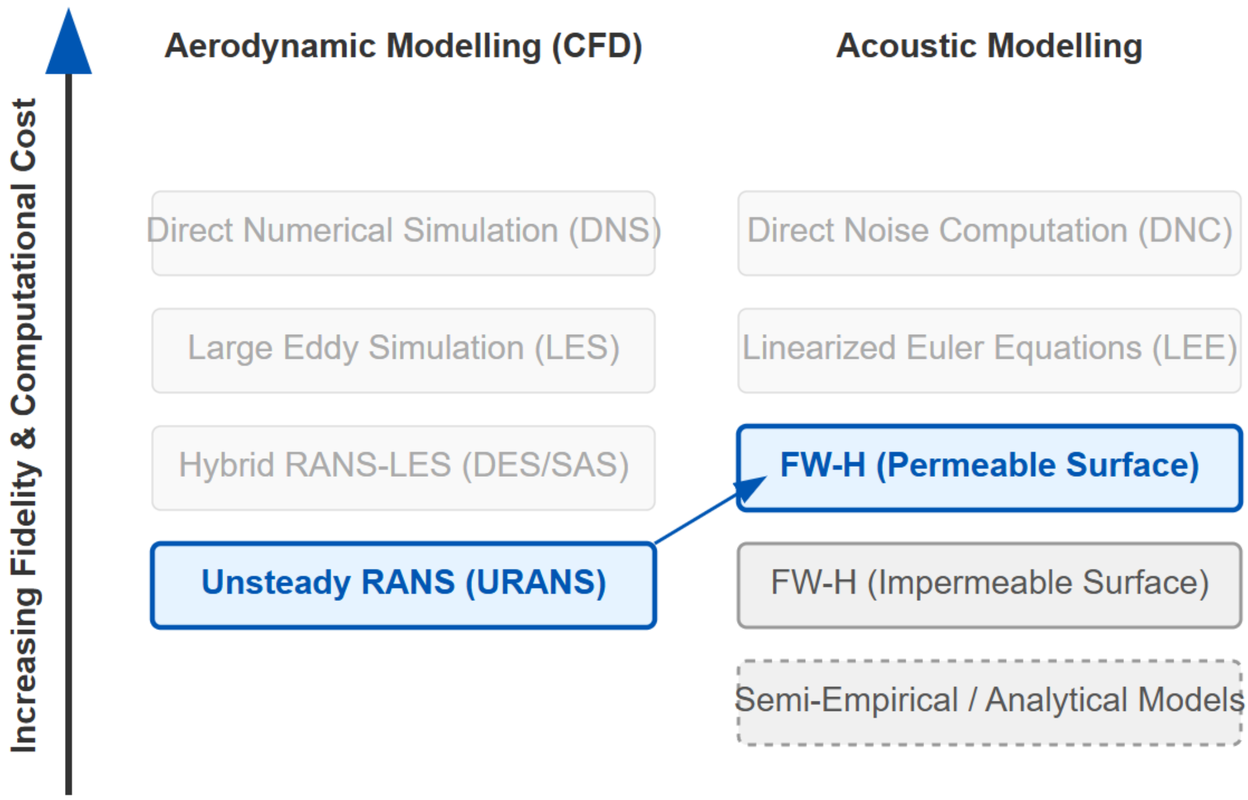


Figure 1.1: Hybrid modelling approach schematic

In this work, the FW-H method is validated and verified for its sensitivity using theoretical point sources, as illustrated in Figure 1.2, to build confidence in its capabilities; the detailed description of which will be provided in the next chapter. After a confidence in the tool has been established, CFD will be used as an input to the FW-H code to form a hybrid computational-hybrid method [6, 11, 17, 18], which replaces the point source in the illustration below with the CFD domain. As the interest in this work is to predict the tonal noise generated by the open rotor, the CFD framework used is the Unsteady Reynolds Averaged

Navier-Stokes (URANS), as it is considered sufficient in providing a good balance between accuracy and computational cost for this purpose [15, 19]. Should a future interest arises in predicting broadband noise, other frameworks such as Detached Eddy Simulation (DES) [20] or Large Eddy Simulations (LES) can be implemented alongside the validated FW-H method.

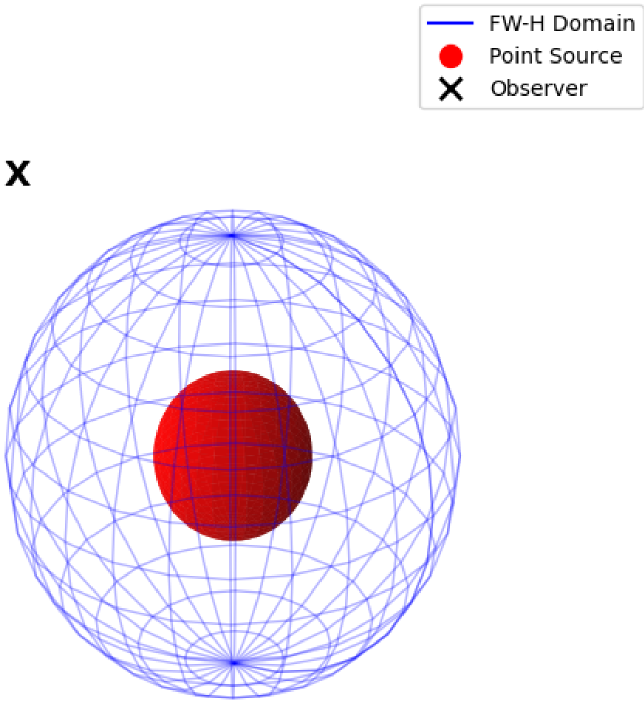


Figure 1.2: Hybrid domain illustration

1.4 Research Objectives and Outlines

The objective of this work are:

1. **To develop an acoustics prediction method using the permeable surface form of the FW-H equation, then validating it using fundamental sound sources**

   This part of the work is carried out in Chapter 2. The background of the permeable surface form of the Ffowcs Williams-Hawkings method is outlined, and the definition of the analytical thickness and loading noise sources are provided as a baseline for the validation methodology. The validation of the numerical FW-H method is performed using the analytical sound sources using test cases that is described within the chapter.

2. **To develop an understanding of the sensitivity of the FW-H method towards the temporal and spatial discretisations**

   This is carried out in Chapter 3.

3. **To use the FW-H method to predict open rotor noise using an appropriate CFD data**

   This is done in Chapter 4

Chapter 2

Background and Validation of the FW-H Method

In this chapter, the FW-H method as well as the analytical methods by which it is validated are described. The formulation used in the numerical FW-H method is described in Section 2.1 below along with the description of how the computer code works. The following sections will then describe the test cases used to validate the code; describe the analytical solutions of the monopole and dipole used to compare against the numerical solution; and finally, the validation data is presented to build confidence in the numerical code itself.

2.1 Description of the FW-H Method

2.1.1 Derivation of the FW-H Equation

To predict the open rotor noise, the FW-H equation in its permeable surface form provides a useful basis for noise prediction. This formulation is based on the governing laws of fluid motion, and the descriptions provided below are based on the original paper by Ffowcs Williams and Hawkings [8]. Consider a control surface *S*, in which contains a combination of fluid and solid boundaries, while the region outside contains only fluid. *S* moves with a velocity v and is defined by the equation $f(x,t) = 0$, the following relationship then hold

$$
\begin{gathered}
f < 0, H(f) = 0 \\
f > 0, H(f) = 1 \\
\nabla_x f = |\nabla_x f|\mathbf{n}, \mathbf{n} \text{ is the outward normal to } S \\
D_v f/Dt = 0, \text{ so } \partial f/\partial t = -v_{\mathbf{n}}|\nabla_x f| \\
\partial H(f)/\partial x_i = \delta(f)\partial f/\partial x_i = \delta(f)\mathbf{n}_i|\nabla_x f|
\end{gathered}
\tag{2.1}
$$

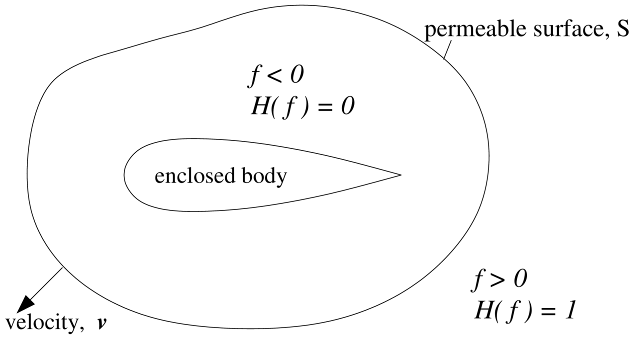


Figure 2.1: Permeable control surface

Generalised flow variables are denoted by an overbar. Outside of the surface, the variables are equal to the real flow variables, while inside they are assigned the value zero, hence are discontinuous acorss *S*. Manipulating the mass and momentum conservation equations,

$$\begin{aligned} H(f)\left(\frac{\partial \rho'}{\partial t} + \frac{\partial \rho u_i}{\partial x_i}\right) &= 0 \\ \frac{\partial \overline{\rho'}}{\partial t} + \frac{\partial (\overline{\rho u_i})}{\partial x_i} &= (\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}}))\delta(f)|\nabla_x f| \end{aligned} \tag{2.2}$$

$$\begin{aligned} H(f)\left(\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j}(p_{ij} + \rho u_i u_j)\right) &= 0 \\ \frac{\partial (\overline{\rho u_i})}{\partial t} + \frac{\partial}{\partial x_j}(\overline{p_{ij}} + \overline{\rho u_i u_j}) &= (p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}}))\delta(f)|\nabla_x f| \end{aligned} \tag{2.3}$$

subtracting the divergence 2.3 from the time derivative 2.2 gives an inhomogeneous wave equation,

$$\begin{aligned} \Box^2 p'(x,t) = {} & \frac{\partial^2 \overline{T_{ij}}}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_i}\left(p_{ij}\mathbf{n}_j + \rho u_i(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)|\nabla_x f| \\ & + \frac{\partial}{\partial t}\left(\rho_0 u_{\mathbf{n}} + \rho'(u_{\mathbf{n}} - v_{\mathbf{n}})\right)\delta(f)|\nabla_x f| \end{aligned} \tag{2.4}$$

Which is the differential form of the FW-H equation, and is the equation governing the generation and propagation of sound. Converting this differential form into an equivalent integral form is done using Green's function. A coordinate change from the fixed one to a moving one with the control surface η is performed so the sources are at rest in the η space. The Jacobian for the coordinate change is J which represents the ratio of volume elements in the η and y spaces. Performing the steps above gives the inhomogeneous wave equation in the integral form of the FW-H equation,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = & \frac{\partial^2}{\partial x_i \partial x_j} \int_{-\infty}^{\infty} \frac{\overline{T_{ij}}\delta(g)J}{4\pi r} d^3\eta d\tau \\
& - \frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{(p_{ij}\mathbf{n}_j + \rho u_i(u_\mathbf{n} - v_\mathbf{n}))\,\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau \\
& + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{(\rho_0 u_\mathbf{n} + \rho'(u_\mathbf{n} - v_\mathbf{n}))\,\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau
\end{aligned} \tag{2.5}
$$

The first RHS term, the quadrupole, is not dealt with explicitly [21, 3] as it is more significant for transonic and supersonic Mach numbers [8, 22]. By choosing a sufficiently large permeable surface, the quadrupole contributions outside of this surface is negligible,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = & -\frac{\partial}{\partial x_i} \int_{-\infty}^{\infty} \frac{(p_{ij}\mathbf{n}_j + \rho u_i(u_\mathbf{n} - v_\mathbf{n}))\,\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau \\
& + \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \frac{(\rho_0 u_\mathbf{n} + \rho'(u_\mathbf{n} - v_\mathbf{n}))\,\delta(f)\delta(g)|\nabla_x f|J}{4\pi r} d^3\eta d\tau
\end{aligned} \tag{2.6}
$$

Using the notation by di Francescantonio [23] to simplify the algebra, new variables U_i and L_i are introduced to represent mass and momentum fluxes through the control surface,

$$
U_i = (1 - \frac{\rho}{\rho_0})v_i + \frac{\rho}{\rho_0}u_i \quad L_i = p_{ij}\mathbf{n}_j + pu_i(u_\mathbf{n} - v_\mathbf{n})
$$

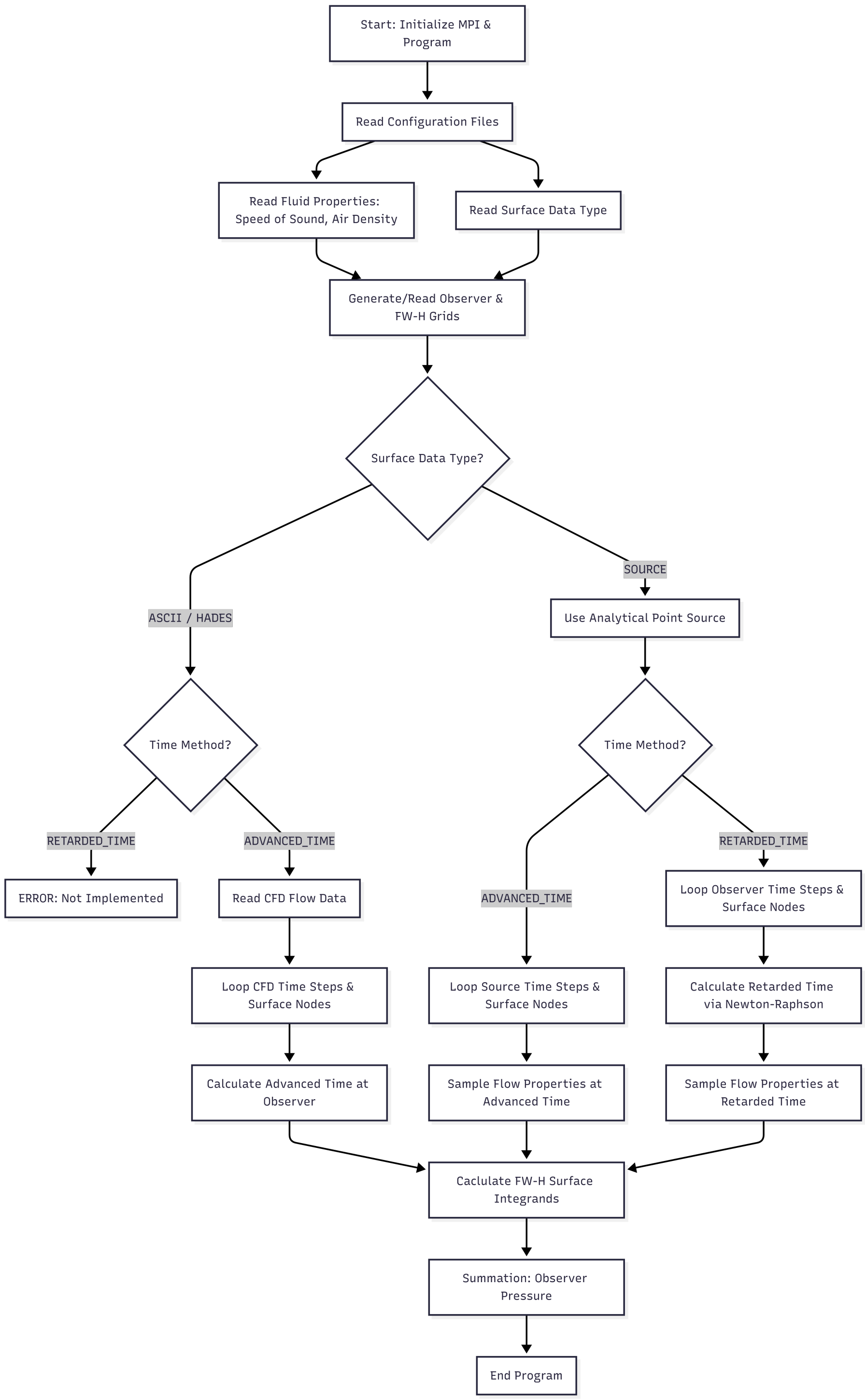
Considering an undistorted control surface that translates at a constant and subsonic velocity, the retarded time formulation, as described by Brentner and Farassat [24] is,

$$
\begin{aligned}
p'(\boldsymbol{x},t) = & \frac{1}{4\pi} \int_S \left[\frac{\rho_0(\dot{U}_\mathbf{n} + U_{\hat{\mathbf{n}}})}{r(1-M_r)^2} \right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi} \int_S \left[\frac{\rho_0(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1-M_r)^3} \right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi c} \int_S \left[\frac{\dot{L}_r}{r(1-M_r)^2} \right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi} \int_S \left[\frac{L_r - L_m}{r^2(1-M_r)^2} \right]_{\tau^*} dS(\eta) \\
& + \frac{1}{4\pi c} \int_S \left[\frac{L_r(r(dM/d\tau)_r + c(M_r - |M|^2))}{r^2(1-M_r)^3} \right]_{\tau^*} dS(\eta)
\end{aligned} \tag{2.7}
$$

The retarded time are calculated implicitly via the Newton-Raphson root-finding method.

2.1.2 Description of the FW-H Code

The FW-H code was written in Fortran 90 to implement the calculations computationally using equation 2.7. The program is summarised in the figure below,



The ambient flow properties are first defined to be used as reference values. The code then checks whether the input is 'Source' or 'ASCII/HADES'; the former uses the analytical point source that will be described in the following sections, and the latter uses CFD data as input and uses its grid

2.2 Validation of the FW-H Code

To compare the accuracy of the derivations described in Chapter **??**, several test cases are used to model a point sound source and an observer. This section describes the test cases used and the validations. Figure 2.2 below illustrates the framework used for validation.

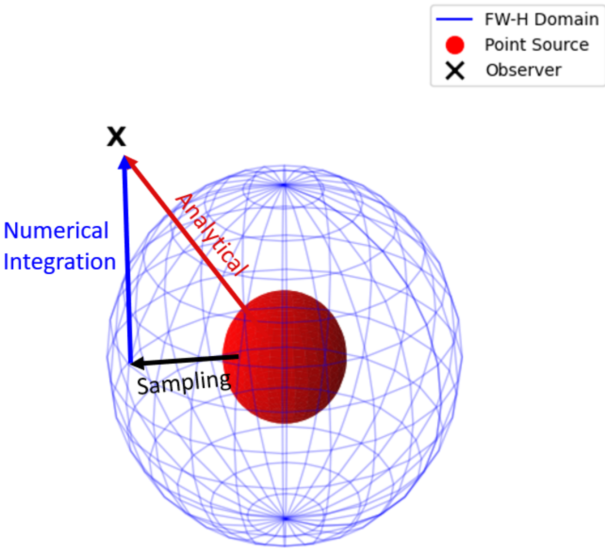


Figure 2.2: Code validation framework

The numerical code samples the source to the permeable boundary (black arrow), followed by surface integration of pressure to calculate the acoustic pressure at the observer (blue arrow). Analytical pressure is calculated directly from the source to the observer (red arrow).

2.2.1 Test Cases

In all validation test cases, the point source is prescribed with strength $Q(\tau) = \sin(10\tau)$, from a starting point at the origin, $[0, 0, 0]$. Speed of sound c and freestream density ρ_0 are set to $340m/s$ and $1.2kg/m^3$, respectively. The observer's location depends on the case. A control surface surrounding the source are prescribed with a radius r.

These following are based on the work of Char [25] and are extended for dipoles:

- Stationary and co-translating point source and observer. This test case is used as a baseline to test the overall prediction of the zero relative motion between point source and observer. The tests will also demonstrate the similarity between the two cases.

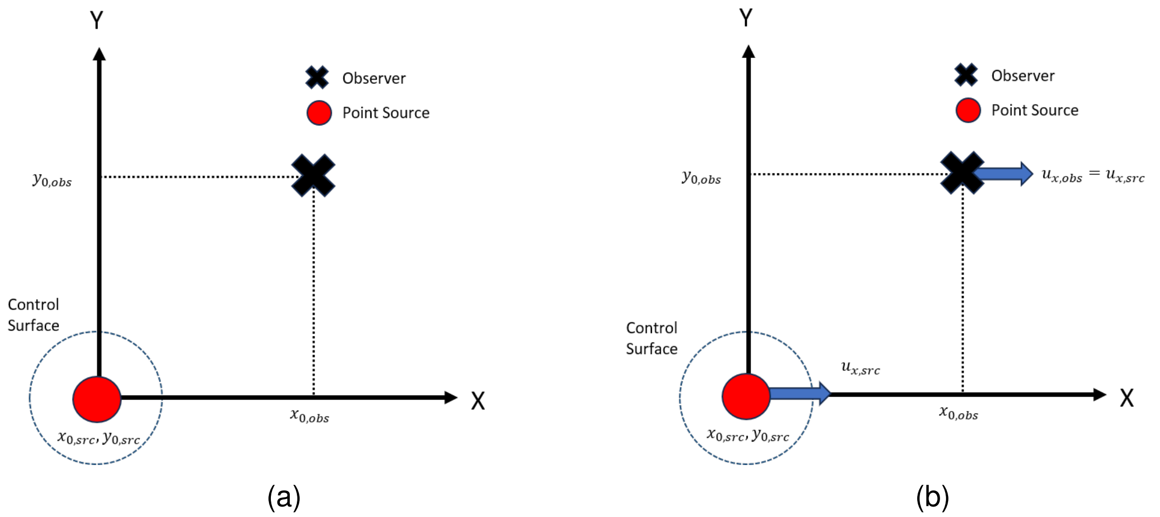


Figure 2.3: Illustration of, a) stationary point source and observer, and b) co-translating point source and observer cases

- Translating point source and stationary observer. This test case demonstrates the prediction with cases of non-zero relative motion between point source and observer, and thus shows the limitations of some of the derivations.

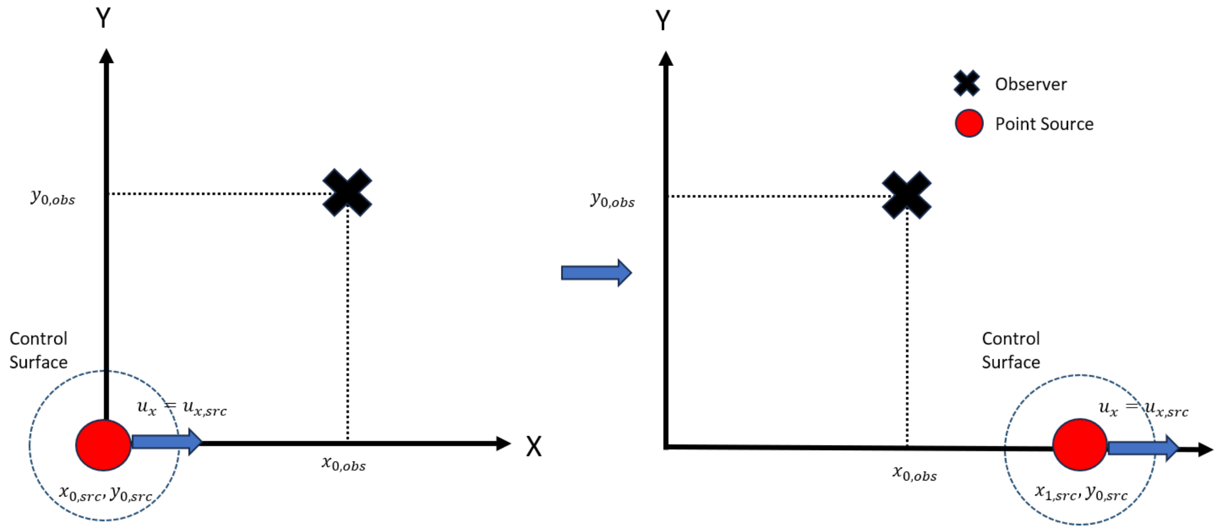


Figure 2.4: Illustration of translating point source and stationary observer cases

Any error are then be quantified using the following,

$$Error = \frac{\Sigma_j (p_{j,ref} - p_{j,num})^2}{\Sigma_j (p^j_{ref})^2} \tag{2.8}$$

2.3 Analytical Solution for a Monopole

Analytical approaches for monopole sources have been validated in previous work by Char and Thomas [25, 17]; however, expansion into dipoles requires further validation. Monopoles or thickness sources are sounds that are equivalent to mass supply, described as the following,

$$\Box^2 p'(\boldsymbol{x}, t) = \frac{\partial}{\partial t}(Q(t)\delta(\boldsymbol{x} - y(t))) \tag{2.9}$$

Convolving the above with the Green's function in 3D free-space, described as,

$$G = \frac{\delta(g(\tau))}{4\pi r} \tag{2.10}$$

where $g(\tau) = t - \tau - \frac{r(\tau)}{c}$ and $r(\tau) = |\boldsymbol{x} - y(\tau)|$, gives the following,

$$p'(\boldsymbol{x}, t) = \frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r(1 - M_r)}\right]_{\tau *} \tag{2.11}$$

which is the description of the monopole used in this work.

2.3.1 Monopole Pressure Equation

Taking the monopole definition as described in Equation 2.11, taking the differentiation using the definitions as described in Appendix B, and convolving the analytical solution for pressure is obtained,

$$p'(\boldsymbol{x}, t) = \frac{1}{4\pi}\left[\frac{\partial Q(\tau)/\partial \tau}{r(1 - M_r)^2} + \left(\frac{\partial M}{\partial \tau}\right)\frac{Q(\tau)}{r(1 - M_r)^3} + \frac{Q(\tau)c(M_r - |M|^2)}{r^2(1 - M_r)^3}\right]_{\tau *} \tag{2.12}$$

2.3.2 Monopole Velocity Equation

Using the momentum equation,

$$\rho_0 \frac{\partial u_i}{\partial t} + \rho_0\left(u_i \frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right) \tag{2.13}$$

Assuming the flow has no mean flow and is inviscid, substituting the monopole definition into the pressure term, and convolving with Green's function to replace $\partial/\partial x_i$ with $\partial/\partial t$,

$$u_i' = \frac{1}{4\pi\rho_0}\left[\frac{(\partial Q(\tau)/\partial\tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau}\right)\frac{Q(\tau)r_i}{cr^2(1-M_r)^3} - \frac{Q(\tau)M^2r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2}\right]_{\tau*} \tag{2.14}$$

2.3.3 Monopole Density Equation

The expression for density is the same for both monopole and dipole since the flow is assumed to be linear and isentropic. Ttherefore, the expression for density is,

$$\rho' = \frac{p'}{c^2} \tag{2.15}$$

2.4 Validation of the FW-H Method for Monopole Sources

The first investigation are of monopole with stationary, as well as co-translating source and observer. This gives a baseline sensitivity of different control surface grid resolutions and time step sizing. The radius of the control surface is set to $1m$, and the observer is located at $[50, 50, 0]$. In the the co-translation case, the points are translating at $150m/s$ along the x-direction.

For grid sensitivity, the control surface is discretised with 40×40, 15×15, and 5×5 grids and the time step size is set to $dt = 0.005s$. Conversely, for the time step sizing test, the grids are set to 40×40 resolution while the time steps are sized to $dt = 0.005s$, $dt = 0.025s$, and $dt = 0.125s$, which correspond to $200Hz$, $40H$z, and $8Hz$ sampling frequencies, respectively.

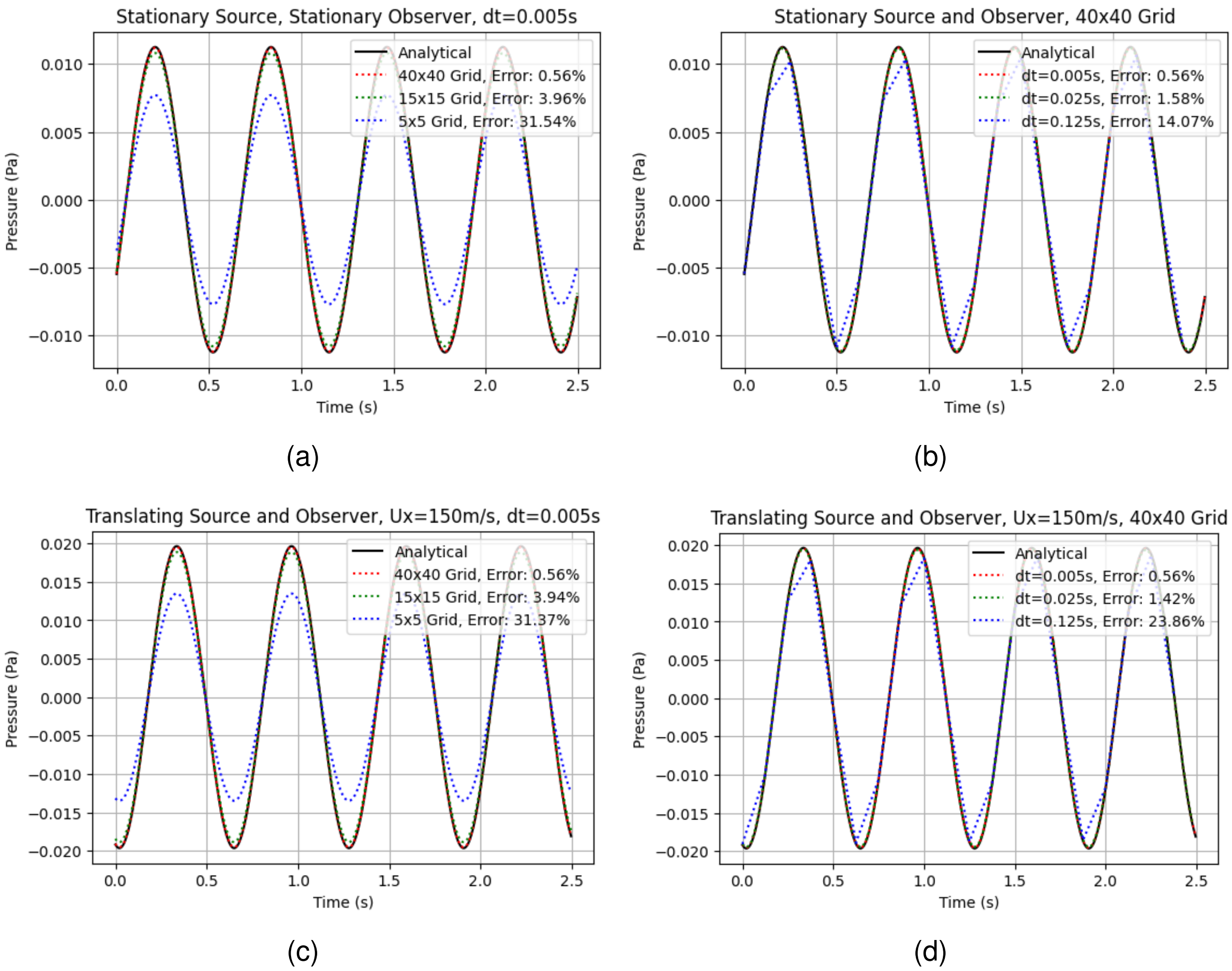


Figure 2.5: Stationary and co-translating monopole test cases, a) stationary with varied grid sizing, and b) stationary with varied time-stepping sizing, c) co-translating with varied grid sizing, and d) co-translating with varied time-stepping sizing

For the relative motion case, the grid and time-stepping sensitivity set ups remain the same. The cases differ by translating only the source at $150m/s$ along the x-axis, while the observer remains stationary.

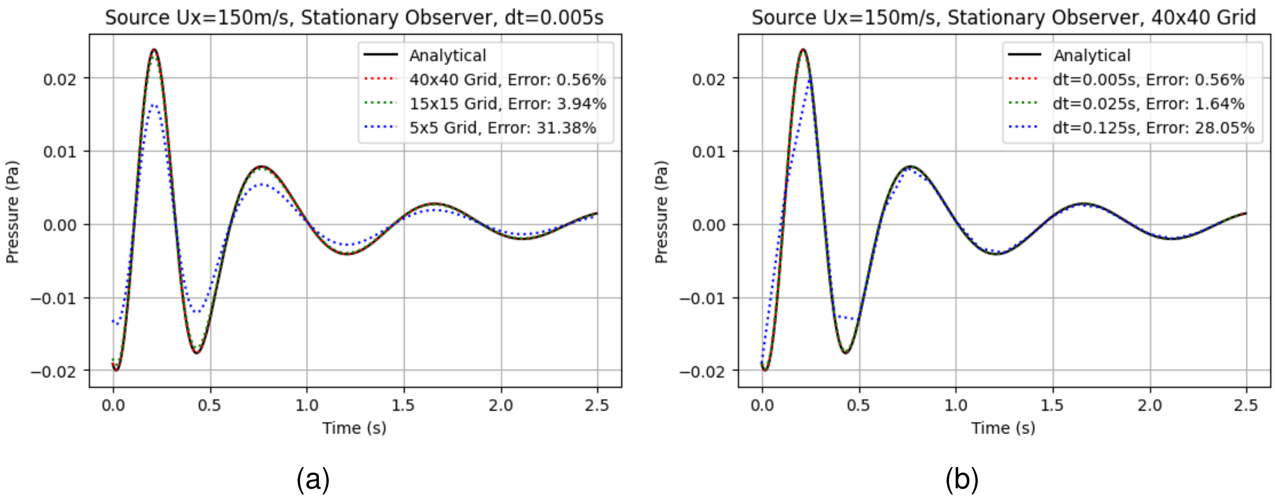


Figure 2.6: Translating monopole with stationary observer test case, a) with varied grid sizing, and b) with varied time-step sizing

Comparing the first two cases (Figure 2.5) shows a distinction in amplitude and phase, despite both scenarios having zero relative Mach number ($M_r = 0$). In the co-translating case, the source moves through the quiescent fluid medium, which causes a phase shift of $\sim \pi/2$ and pressure increase due to the convective term. Furthermore, the emission distance r in the retarded time, $\tau = t - r/c$, differs from the geometric distance because both points are moving through the medium during the propagation time. The third case demonstrates the effect of relative Mach number. In Figure 2.6 the Doppler effect causes the acoustic frequency and pressure amplitude received by the observer to increase as the source approaches ($0 \leq t \leq 0.33s$), and decrease as it moves away ($t > 0.33s$).

In the grid resolution comparisons, while the finest resolution of 40×40 has very good agreement with the analytical solutions at less than 1% error in all cases, the errors progressively increase to $\sim 4\%$ and $\sim 30\%$, for coarser resolutions. The plots show at lower resolution, the numerical results could not capture the peaks and troughs of the harmonic wave, causing dissipation. Given the acoustic wavelength is $\sim 214m$ (from the prescribed frequency and speed of sound values), the coarsest grid has an arc distance between two nodes of 1.25 meters. The errors shown in the grid sensitivities are mainly due to geometric error of the spherical domain, rather than the grids not being able to capture the wavelengths. This shows that this grid resolution is sufficient given the input and control surface radius.

The time step sensitivity study exhibits similar trends. The $dt = 0.005s$ and $dt = 0.025s$ sizes yield errors at less than 1% and 2%, respectively. However, largest time-step size of $dt = 0.125s$ (an $8Hz$ sampling frequency), is insufficient to capture the input acoustic frequency of $10rad/s$ or $\sim 1.6Hz$, as it can only sample at five times the input. This can be seen in the lags in the plots as the numerical solution attempted to keep up with the wave.

2.5 Analytical Solution for a Dipole

Dipoles or loading sources are sounds due to forces acting on the fluid, equivalent to body forces and momentum supply. It is also equivalent to two equal monopoles with opposite signs at an infinitesimally small distance. The dipole is described as the following,

$$p'(\boldsymbol{x},t) = -\frac{\partial}{\partial x_i}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau *} \tag{2.16}$$

In the dipole description above, the source forcing term f_i is not an arbitrary vector. To analytically model a dipole, the mathematical equivalence between monopoles and dipoles must be established. Sound pressure generated by a monopole source is defined as,

$$p'_{mono}(\boldsymbol{x},t) = \frac{\partial}{\partial t}\left(\frac{Q(\tau)}{4\pi r|1-M_r|}\right)$$

By describing a dipole as two monopoles of equal strength and opposite signs separated by an infinitely small distance δ, the dipole pressure is a superposition of the two monopoles,

$$p'_{dipole} = p'_{mono}(\boldsymbol{x},t) - p'_{mono}(\boldsymbol{x}-\delta,t)$$

As δ tends to zero, this distance can be expressed as a first-order Taylor series expansion,

$$p'_{dipole}(\boldsymbol{x},t) \approx -\delta\frac{\partial}{\partial x_i}\frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r|1-M_r|}\right]$$

Hence, the dipole source strength $f_i(t)$ can be defined as

$$f_i(t) = 4\pi r|1-M_r|\delta\frac{\partial}{\partial t}\left(\frac{Q(t)}{4\pi r|1-M_r}\right) \tag{2.17}$$

By defininig the force vector in this form and evaluating it via central finite differences, the numerical solver captures the convective derivative $\partial M_r/\partial t$ without requiring an analytical expansion.

To analytically derive the velocity for dipole, explicitly integrating the momentum equation is unviable, as it will return too many terms. Instead, limited solutions are obtained by applying some physical assumptions. The velocity derivations assume either a zero relative Mach number or a plane wave condition. Alternatively, another method that uses the superposition of two equal monopoles of opposite signs will also be presented.

2.5.1 Dipole Pressure Equation

Using the definition described in Equation 2.16, replacing the $\partial/\partial x_i$ term as outlined in Appendix D, and using the steps from monopole pressure, the pressure for a dipole is obtained,

$$\begin{aligned} p'(\boldsymbol{x},t) = & \frac{(\partial f_i(\tau)/\partial\tau)r_i}{4\pi r^2 c(1-M_r)^2} + \frac{f_i(\tau)r_i}{4\pi r^3|1-M_r|^2} - \frac{f_i(\tau)M_r}{4\pi r^2|1-M_r|^2} \\ & + \left(\frac{\partial M}{\partial\tau}\right)\frac{f_i(\tau)r_i}{4\pi r^2 c|1-M_r|^3} + \frac{f_i(\tau)r_i(|M|^2-|M_r|^2)}{4\pi r^3|1-M_r|^3} \end{aligned} \tag{2.18}$$

2.5.2 Dipole Velocity Equation: Zero Relative Mach Number Assumption

Assuming inviscid and no mean flow in the momentum equation, the pressure gradient is,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau*}$$

Hence, the velocity is defined as,

$$u_i' = \frac{1}{4\pi\rho_0}\frac{\partial^2}{\partial x_i\partial x_j}\int_{-\infty}^{t}\frac{f_i(\tau)}{r(1-M_r)}dt$$

substituting the integrand to retarded time and assuming zero relative velocity, thus $dr/dt = 0$ and $M_r = 0$, letting $F_i(\tau) = \int f_i(\tau)d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, the analytical solution for dipole velocity that is valid for cases with zero relative Mach number is obtained,

$$u_i' = \frac{1}{4\pi\rho_0}\left(\frac{F_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{r^3} + \frac{f_j(3\mathbf{n}_i\mathbf{n}_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(\mathbf{n}_i\mathbf{n}_j)}{c^2 r}\right) \tag{2.19}$$

2.5.3 Dipole Velocity Equation: Plane Wave Assumption

The plane wave assumption allows relative Mach number between the source and observer, provided the control surface is within the far-field of the source. The far-field is defined as,

$$k\hat{r} \gg 1 \tag{2.20}$$

Physically, this means the acoustic wavelength is smaller than the control surface diameter, allowing the entire wavelength to be fully captured by the permeable sphere. Using the derivations in Appendix G, the velocity term with the plane wave assumption is obtained,

$$u_i' = \frac{\mathbf{n}_i}{\rho_0 c} p'(\boldsymbol{x}, t) \tag{2.21}$$

2.5.4 Dual Monopole

An alternative approach to validate dipole sources is the dual monopole method. Using the definition of a dipole, this method uses two equal monopoles of opposing signs, separated by a small distance δ, which in this case is defined as,

$$k\delta \ll 1 \tag{2.22}$$

To simplify the implementation, the analytical code calculates $\delta = 0.01/k$. Another advantage of this approach is that it allows the prediction of the directivity patterns of the dipole.

2.6 Validation of the FW-H Method for Dipole Sources

With the FW-H numerical solver validated for monopoles, the validation is extended to dipole sources. As mentioned in Subchapter 2.5, explicitly deriving the velocity terms for dipole sources is unviable. This led to two derivation cases with physical assumptions and one using the two monopoles method. Therefore, the validation is separated into two parts; one with the dipole derivations, and another with the dual monopole under the same conditions. Then, the numerical results are compared to understand the limitations of each approach.

2.6.1 Dipole Derivation Validations

Unlike the monopole validation, the co-translating test case is omitted for the dipole cases to isolate the effects of a non-zero relative mach number. Therefore, the observer is stationary in all test cases. For all subsequent dipole test cases, the dipole axis and translation velocity are strictly imposed along the x-axis, while the observer maintains a transverse offset.

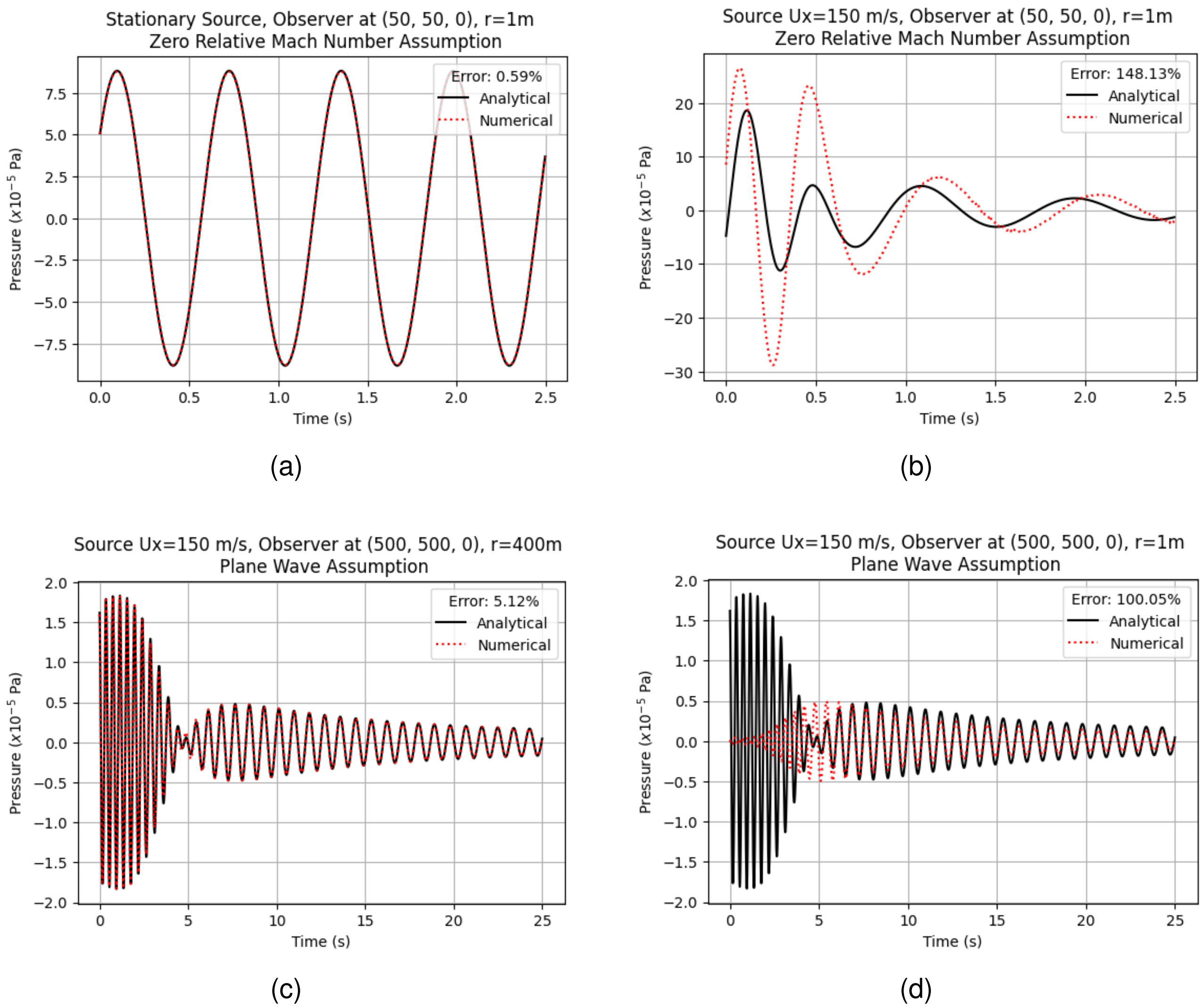
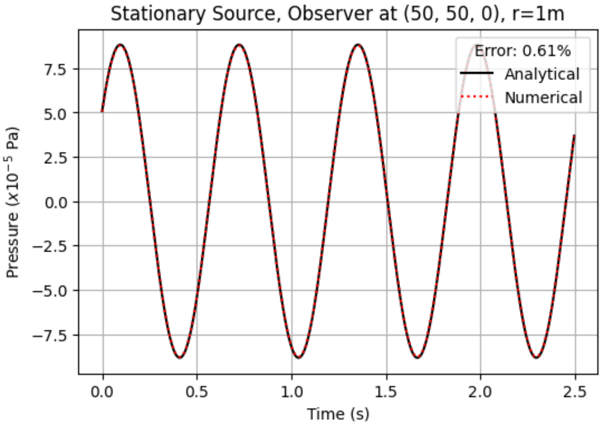


Figure 2.7: Comparison of analytical and numerical dipole. Top: zero relative Mach number case, with a) following, and b) violating the assumption. Bottom: plane wave case, with c) following, and d) violating the assumption.

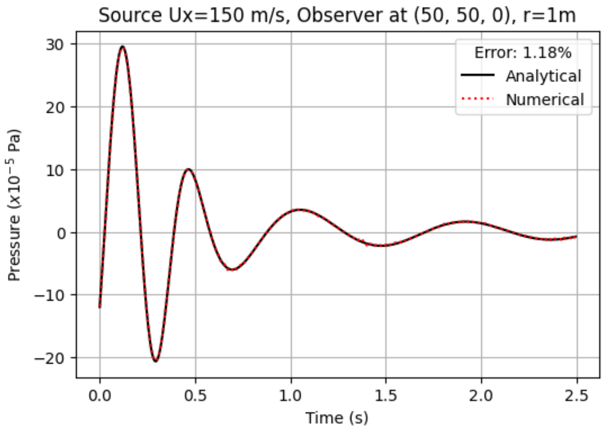
For the zero relative Mach number assumption, the tests examine how relative motion affect the results. In each case, the observer is put at $[50, 50, 0]$. To examine the integration accuracy, the first case examines a stationary problem, and the second one a relative motion problem, moving the source at $150m/s$ along the x-direction. In the plane wave assumption, the tests only examine the relative motion and putting the observer at $[500, 500, 0]$, to be in the "far-field." The control surface radius is expanded to $r = 400m$ in the valid case and to $r = 1m$ in the invalid case.

For stationary source and true far-field observer (Figures 2.7a and 2.7c), the results have good agreements at $< 6\%$ error. However, introducing translation into the former and setting near-field surface to the latter violate the assumptions. For the $M_r = 0$ case, the errors are due to the non-existent relative motion in Equation 2.19; for the plane wave assumption, the control surface being in the near-field of the source. The agreement as the source moves away from the observer is due to the observer eventually being in the physical far-field.

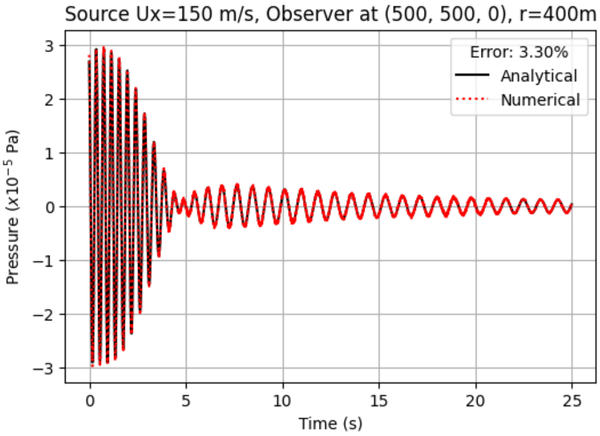
2.6.2 Dual Monopole Validations



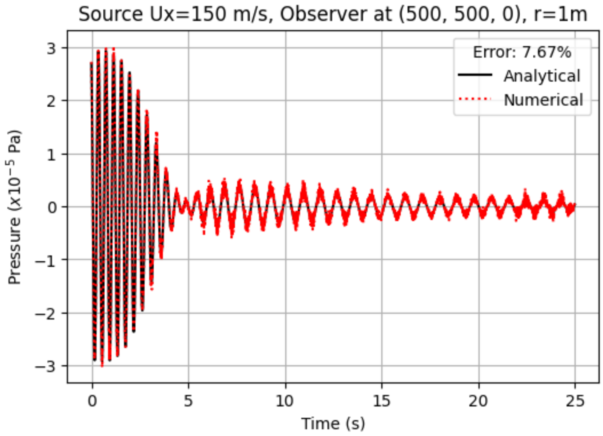
(a)



(b)



(c)



(d)

Figure 2.8: Comparison of analytical and numerical dipole validations using the dual monopole method. Top: zero relative Mach number test cases, and bottom: plane wave test cases, the numerical results for all cases agree very well with the analytical solution.

In contrast to the analytical derivations, the dual-monopole method is significantly more robust in all cases, with highest error at $< 8\%$. By superimposing two monopoles separated by $k\delta \ll 1$, the solver retains the Doppler factor, which is omitted by the zero relative motion assumption. Similarly for the bottom figures, as the monopole was validated using a control surface in the near-field, the results generally agree in the cases which the far-field assumption failed when using the same control surface. It is noted that in Figure 2.8d, the result exhibits more significant errors as the source moves away. This could be attributed to the limitations in the solver and will be discussed in the next chapter.

2.6.3 Numerical Result Comparisons for Dipole Derivations and Dual Monopoles

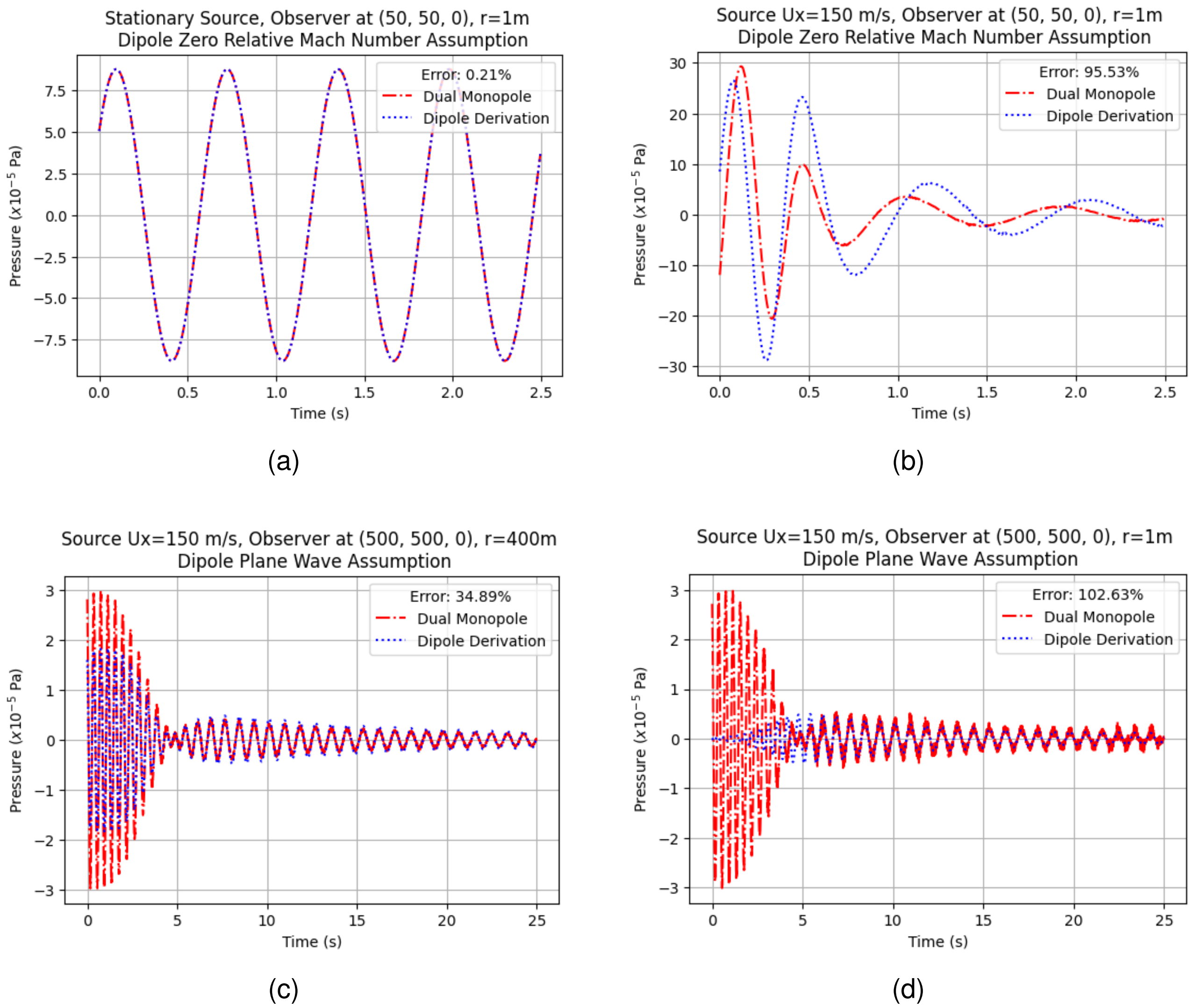


Figure 2.9: Comparison of numerical dipole validations. Top: zero relative Mach number cases, and Bottom: plane wave cases. Left column are cases where the dipole assumptions followed the analytical solutions show good agreement with dual monopole. Right column shows where the dipole assumptions are violated.

Comparing the dipole derivations to the dual monopole approach shows that there are apparent deviations in results other than the stationary case. For Figure 2.9b, it is obvious that the cause is the lack of Doppler effect in the assumption. Figure 2.9c is interesting due to a significant difference in amplitude as the sources approach the observers. Then as they move away, a good agreement in amplitude is observed. This is likely due to the imposing of the x-axis in the dipole source, because as it moves away, the y-axis contribution in the emission distance becomes insignificant.

2.6.4 Directivity

The dual monopole method allows the investigation of directivity patterns of a dipole. Two standard tests are conducted, where the dipole is stationary and another that it moves at $u_x = 10m/s$ to demonstrate the convective effects. The observers are set as a sphere located at a distance of $50m$, from the source.

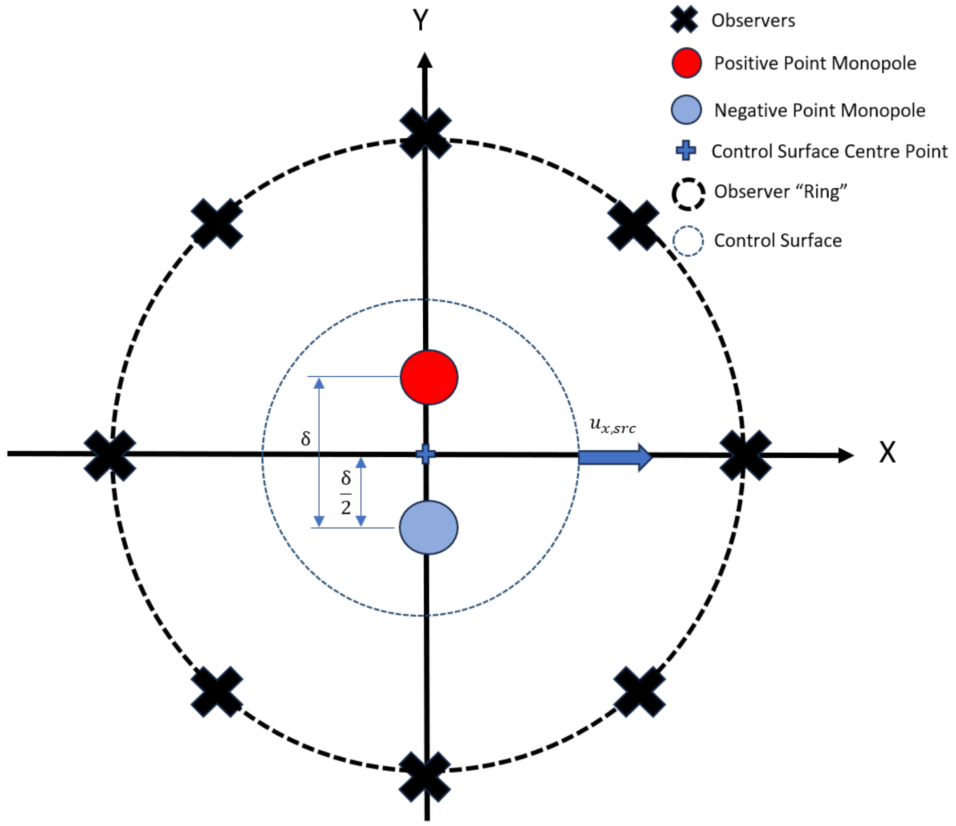


Figure 2.10: Test case for directivity

The stationary case show a standard directivity "figure-8" pattern, showing the locality of the individual monopoles. Imposing a translation to the source alters this pattern towards the x-axis as the source moves in said direction.

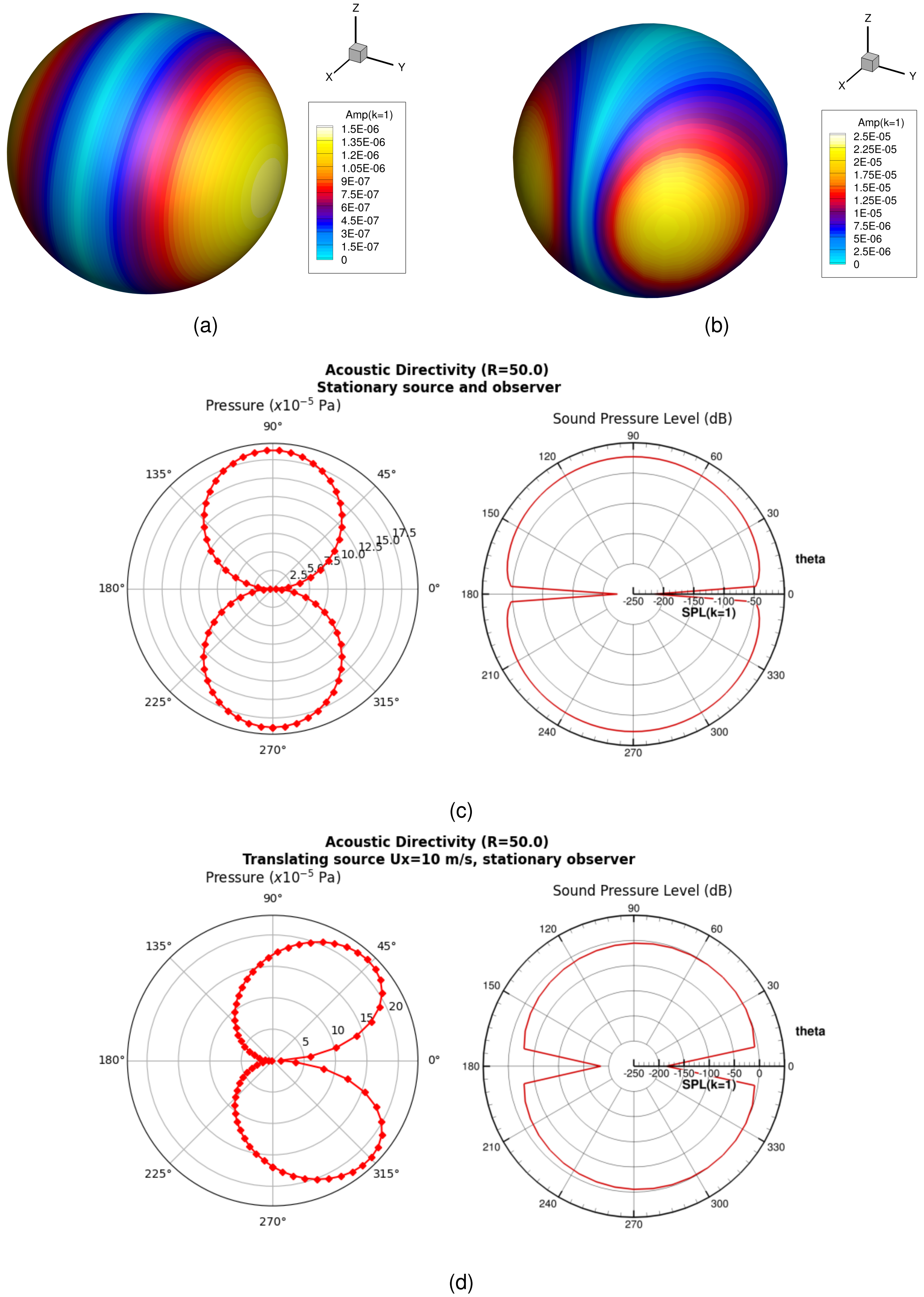


Figure 2.11: Directivity of dual monopoles, stationary case with a) surface plot and c) polar plot; and translating case with b) surface plot and d) polar plot

2.7 Concluding Summary

Chapter 3

Sensitivity Analysis of the FW-H Method

Chapter 4

Open Rotor

Chapter 5

Conclusions

A Ffowcs Williams-Hawkings (FW-H) acoustic code has been analytically validated. The numerical integration is accurate, matching analytical derivations for stationary and moving monopole sources with less than 1% error under sufficient grid and temporal resolution.

For dipole sources, the explicit analytical derivations were shown to break down when subjected to non-zero relative motion or near-field control surfaces. To bypass these limitations, a dual-monopole approach was implemented, proving to be a good test case to demonstrate dipoles. While the numerical integration is mathematically sound, further refinement of the dipole analytical validation method is required. Once complete, the code will be integrated with URANS CFD for open rotor noise prediction.

Chapter 6

Future Work

With the FW-H solver validated with some limitations, the project moves to the hybrid analytical-computational framework. The next part of the work include:

- Validation refinements for dipoles (e.g., using non-stationary monopole as the source term, dynamic vector calculations) to reduce errors.
- Updating the numerical FW-H integration code to use double-precision floating-point arithmetic.
- Development and validation of a baseline URANS-FW-H hybrid model.
- Using the validated URANS-FW-H approach to predict open rotor noise.

Review of Generative AI Use

Generative AI (Google Gemini 3 Pro) has been used in augmenting the research and writing of this report. The usage of which are in helping with LaTeX formatting, formalising some English language uses, mathematical derivations, and checking for programming errors or bugs. The following are prompt examples used on the above main usages:

- ”Help me format this LaTeX document using the structure abstract, table of contents, nomenclature, intro, lit review, derivations, validations, discussion, conclusions, future work, appendices. The word count should only include the main content (from intro to future work).
- ”Help me rephrase these two sentences and bridge between them”
- ”In deriving the velocity equation for the dipole, as the explicit integral of the convolution between the right hand side term and Green’s function is unviable, what would be the implications of assuming that r is a constant value, such that some integrations of the F/(1-M_r) term can be done? If this is solved in the retarded time, the denominator is cancelled, so I can integrate F directly, correct?
- ”As the dual monopole method requires introducing an additional point source, which the code had not considered initially, help me to put an additional point monopole in the sample_properties subroutine in Fortran. As per the derivations, I need to separate the two by a distance delta, which I think can just be calculated as 0.01/k so it dynamically changes if any parameters are changed in the future.

The result of the first two examples being in the formatting of this report and some of the sentences within. None of the result sentences are directly copied from the tool, and instead are rephrased in the author’s own words to reflect the author’s understanding of the topic. The third prompt helped the author in deriving the mathematical parts presented in this report; of which the results are checked and verified via textbooks and during meetings with supervisors. The fourth prompt example returned Fortran codes that are used for the analytical solutions. These solutions tend to come with bugs which the author fixed independently.

In conclusion, all text-based outputs given by the Generative AI were checked, reworded, and edited to be based on the author’s understanding of the topic and to avoid plagiarism. Outputs that are technical are verified by textbooks and discussions with supervisors to ensure the accuracy of the results and limitations of the assumptions.

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Appendix A

Research Project Gantt Chart

The following is the Gantt Chart of the current project up to its completion. Some adjustments had to be made along the duration of the project's execution such as CFD-related parts, however, the overall project objectives and milestones remain the same.

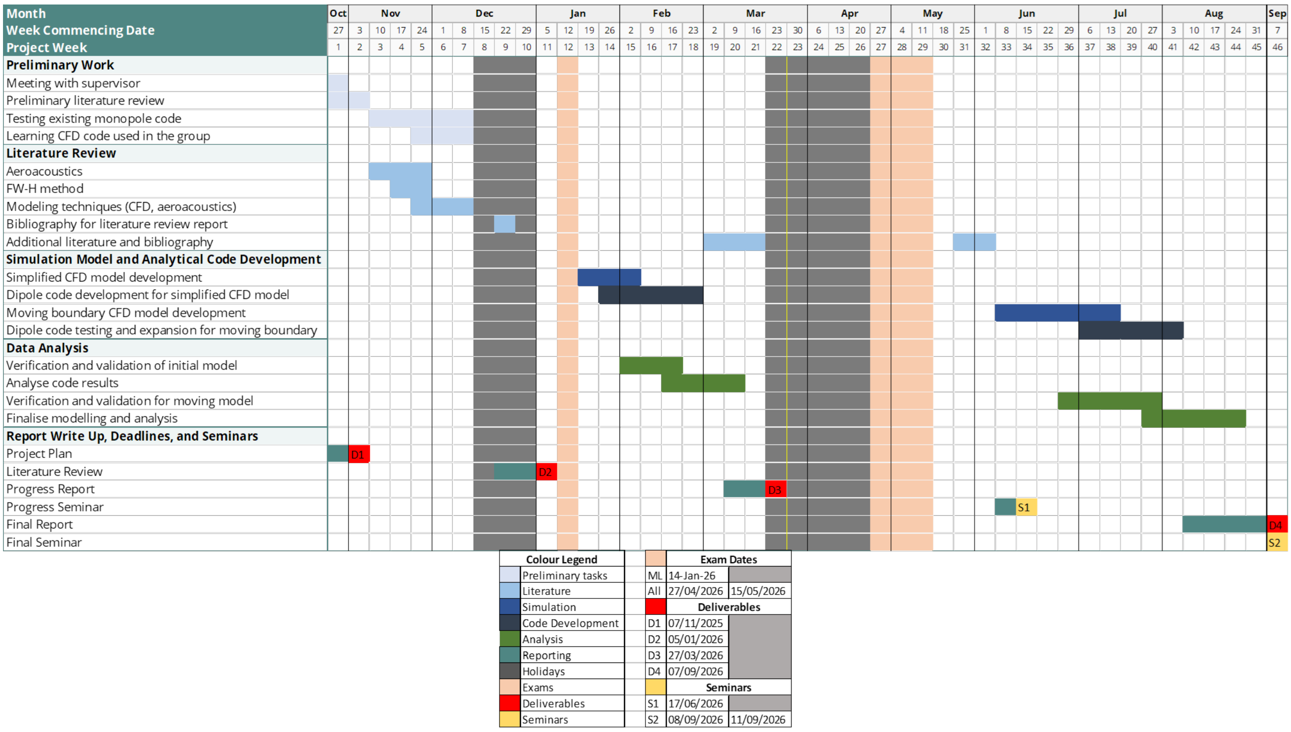

Figure A.1: Gantt Chart of the research project

Appendix B

Variable Derivatives

To derive the analytical point sources, a few definition for the derivatives are established.

$$\left.\frac{\partial r}{\partial \tau}\right|_x = -v_r \tag{B.1}$$

$$\left.\frac{\partial M_r}{\partial \tau}\right|_x = \left(\frac{\partial M}{\partial \tau}\right)_r + \frac{c(M_r^2 - |M|^2)}{r} \tag{B.2}$$

$$\left.\frac{\partial |1-M_r|}{\partial \tau}\right|_x = \frac{(1-M_r)}{|1-M_r|}\left(-\left(\frac{\partial M}{\partial \tau}\right)_r + \frac{c(|M|^2 - M_r^2)}{r}\right) \tag{B.3}$$

$$\left.\frac{\partial [Q(\tau)]_{\tau *}}{\partial \tau}\right|_x = \left.\left[\frac{1}{1-M_r}\frac{\partial Q(\tau)}{\partial \tau}\right]_x\right|_{\tau *} \tag{B.4}$$

$$-\left.\frac{\partial}{\partial x_i}\right|_t \int_S [Q_i]_{\tau *} dS = \left.\frac{\partial}{\partial t}\right|_x \int_S \left[\frac{Q_i r_i}{cr}\right]_{\tau *} dS + \int_S \left[\frac{Q_i r_i}{r^2}\right]_{\tau *} dS \tag{B.5}$$

Appendix C

Green's Function for Wave Equations

Given a wave equation,

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\psi(\hat{r},t) = -f(\hat{r},t)$$

Green's function of the above equation,

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)G(\hat{r},t|\hat{r'},t') = -\delta(\hat{r'}-\hat{r})\delta(t-t')$$

Let $\hat{r'} = 0$ and $t' = 0$. Solving for $\nabla^2 G$

$$\begin{aligned}\nabla^2 G(\hat{r},t) &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)G \\ &= \frac{1}{r}\frac{\partial G}{\partial r} + \frac{1}{r}\left(\frac{\partial G}{\partial r} + r\frac{\partial^2 G}{\partial r^2}\right) \\ &= \frac{1}{r}\frac{\partial^2}{\partial r^2}(rG).\end{aligned}$$

Substituting back to the wave form,

$$\nabla^2 G - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}G = -\delta(r)\delta(t)$$

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(rG) - \frac{1}{c^2}\frac{1}{r}\frac{\partial^2}{\partial t^2}(rG) = -\delta(r)\delta(t)$$

Factoring the differential terms, it is found that if $r > 0$, then the left-hand side term is equal to zero,

$$\left[\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial r}\right)\left(\frac{1}{c}\frac{\partial}{\partial t} - \frac{\partial}{\partial r}\right)\right](rG) = 0 \tag{C.1}$$

Then, let $u = ct - r$ and $v = ct + r$, it will be found that,

$$\frac{\partial}{\partial u} = \frac{1}{2}\left(\frac{1}{c}\frac{\partial}{\partial t} - \frac{\partial}{\partial r}\right) \quad \frac{\partial}{\partial v} = \frac{1}{2}\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial r}\right)$$

Given that $\partial^2(rG)/(\partial u \partial v) = 0$, if $rG = g(v) = g_+(ct + r)$ and $rG = g(u) = g_-(ct - r)$, then,

$$G = \frac{1}{r} g_\pm(ct \pm r)$$
$$G(\hat{r}, t|\hat{r'}, t') = \frac{1}{\hat{r} - \hat{r'}} g_\pm(c(t - t') \pm |\hat{r} - \hat{r'}|)$$

Taking $\nabla^2 G$ using this definition,

$$\nabla^2 G = \nabla^2 \left(\frac{1}{r} g_\pm(ct \pm r)\right) = \nabla^2 \left(\frac{1}{r}\nabla g_\pm + g_\pm \nabla\left(\frac{1}{r}\right)\right)$$
$$= -\frac{2}{r^2}\frac{\partial g}{\partial r} + \frac{2}{r^2}\frac{\partial g}{\partial r} + \frac{1}{r}\frac{\partial^2 g}{\partial r^2} - 4\pi\delta(r)g$$

The first two terms of the right-hand side cancel, and moving back to the wave equation format,

$$\nabla^2 G - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} G = -\frac{1}{r}\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r^2}\right) g_\pm - 4\pi\delta(r) g_\pm \tag{C.2}$$

As described in equation C.1, the first term of the right-hand side cancels. Going back to the wave equation,

$$\Box^2 \psi(\hat{r}, t) = -\delta(\hat{r})\delta(t) = -4\pi\delta(\hat{r}) g_\pm$$

Therefore,

$$\therefore \quad G_\pm = \frac{\delta(ct \pm r)}{4\pi r} \tag{C.3}$$

Appendix D

Replacing $\partial/\partial x_i$ Terms

As the definition of dipoles are that of the spatial derivative of the forcing term convolved with the Green's function in 3D free-space, it is necessary to replace the spatial derivative terms to that of the temporal one to derive using the definitions described in Appendix B. As the goal is to transform $\mathbf{x} \to \mathbf{X}$ and $t \to \tau$, and the following are true: $\mathbf{x} = \mathbf{X}$ and $\tau = t - \frac{\mathbf{r}}{c}$, then

$$\left.\frac{\partial}{\partial x_i}\right|_t = \frac{\partial X_j}{\partial x_i}\frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial x_i}\frac{\partial}{\partial \tau} = \delta_{ij}\frac{\partial}{\partial X_j} - \frac{r_i}{cr(1-M_r)}\frac{\partial}{\partial \tau}$$

$$\frac{\partial}{\partial x_i} = \frac{\partial}{\partial x_i} - \frac{r_i}{cr(1-M_r)}\frac{\partial}{\partial \tau} \tag{D.1}$$

Similarly for the time derivative,

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{x}} = \frac{\partial X_j}{\partial t}\frac{\partial}{\partial X_j} + \frac{\partial \tau}{\partial t}\frac{\partial}{\partial \tau}$$

Assuming no mean flow, the first term of the right hand side, $\partial X_i/\partial t$ is equal to zero, leaving,

$$\left.\frac{\partial}{\partial t}\right|_{\mathbf{x}} = \frac{1}{1-M_r}\left.\frac{\partial}{\partial \tau}\right|_{\mathbf{x}} \tag{D.2}$$

Appendix E

Monopole Velocity Derivations

Using the momentum equation, assuming the flow has no mean flow and is inviscid, substituting the monopole definition into the pressure term, the second terms of both left- and right-hand are zero. Defining the pressure gradient by defining the pressure as monopole pressure definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial t}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau *}$$

and cancelling the unsteady term from both sides gives the following for the momentum equation,

$$u_i = \frac{1}{\rho_0}\frac{\partial}{\partial x_i}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau *} \tag{E.1}$$

Convolving with Green's function in free 3D-space is done to replace the spatial derivative into a temporal derivative,

$$p'(\boldsymbol{x},t) = \frac{\partial}{\partial x_i}(Q_i \star G) = -\frac{\partial}{\partial t}\left(Q_i \star \frac{x_i G}{c|\boldsymbol{x}|}\right) - Q_i \star \frac{x_i G}{|\boldsymbol{x}|^2}$$

which has the same form as the integral shown in Appendix B, Equation B.5. Substituting the monopole source terms into Q_i, and converting to retarded time using the equation found in Appendix D Equation D.2, gives,

$$\frac{\partial}{\partial x_i}\left[\frac{Q(\tau)}{4\pi r(1-M_r)}\right]_{\tau *} = \frac{1}{1-M_r}\frac{\partial}{\partial \tau}\left[\frac{Q(\tau)r_i}{4\pi c\boldsymbol{r}^2(1-M_r)^2}\right]_{\tau *} + \left[\frac{Q(\tau)r_i}{4\pi r^3(1-M_r)}\right]_{\tau *}$$

which can then be solved analytically to give the solution for velocity,

$$u_i = \frac{1}{4\pi\rho_0}\left[\frac{(\partial Q(\tau)/\partial\tau)r_i}{cr^2(1-M_r)^2} + \frac{Q(\tau)r_i}{r^3(1-M_r)^3} + \left(\frac{dM}{d\tau}\right)\frac{Q(\tau)r_i}{cr^2(1-M_r)^3} - \frac{Q(\tau)M^2 r_i}{r^3(1-M_r)^3} - \frac{Q(\tau)M_i}{r^2(1-M_r)^2}\right]_{\tau*} \tag{E.2}$$

Appendix F

Dipole Velocity: Zero Relative Mach Number Assumption Derivations

Defining pressure gradient from the momentum equation and dipole definition,

$$-\frac{\partial p}{\partial x_i} = \frac{\partial^2}{\partial x_i \partial x_j}\left[\frac{f_i(\tau)}{4\pi r(1-M_r)}\right]_{\tau *}$$

Going back to momentum equation, the velocity is,

$$u_i = \frac{1}{4\pi\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t}\frac{f_i(\tau)}{r(1-M_r)}dt$$

assuming $dr/dt = 0$ and $M_r = 0$, letting $F_i(\tau) = \int f_i(\tau)d\tau \implies f_i(\tau) = \dot{F}_i(\tau)$, and changing to integrating with respect to τ, this simplifies the integrand to,

$$u_i = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t} f_i(\tau)d\tau$$

Taking the integral,

$$u_i = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\int_{-\infty}^{t} f_i(\tau)d\tau = \frac{1}{4\pi r\rho_0}\frac{\partial^2}{\partial x_i \partial x_j}\left[F_i(\tau - r/c)\right]$$

Hence, the analytical solution for velocity that is valid for cases of zero relative Mach number between the source and observer for near- and far-field,

$$u_i = \frac{1}{4\pi\rho_0}\left(\frac{F_j(3n_in_j - \delta_{ij})}{r^3} + \frac{f_j(3n_in_j - \delta_{ij})}{cr^2} + \frac{\dot{f}_j(n_in_j)}{c^2r}\right) \tag{F.1}$$

Appendix G

Dipole Velocity: Plane Wave Assumption Derivations

Given the plane wave,

$$p'(\boldsymbol{x}, t) = \frac{g(t - r/c)}{r} \tag{G.1}$$

Taking the pressure gradient,

$$\frac{\partial p'}{\partial x_i} = \left[\frac{\partial}{\partial x_i}\frac{1}{r}g\right] + \frac{1}{r}\left[\frac{\partial}{\partial x_i}g\right]$$

where the first and second terms correspond to the near- and far-field, respectively. The plane wave is valid only in the far-field, and taking the time derivative yields,

$$\frac{\partial p'}{\partial t} = \left[\frac{\partial}{\partial t}\frac{g}{r}\right] = \frac{1}{r}\frac{\partial}{\partial t}g(\tau) - g(\tau)\frac{v_r}{r^2}$$

In the far-field, $\frac{1}{r}\frac{\partial}{\partial t}g(\tau) \gg g(\tau)\frac{v_r}{r^2}$, simplifying the expression to,

$$\frac{\partial p'}{\partial t} \approx \frac{1}{r}\frac{\partial}{\partial t}g(\tau)$$

which gives the following expression,

$$\frac{\partial p'}{\partial x_i} = -\frac{n_i}{c}\frac{\partial p'}{\partial t}$$

substituting into the momentum equation provides the velocity equation for the plane wave,

$$u_i = \frac{n_i}{\rho_0 c}p'(\boldsymbol{x}, t) \tag{G.2}$$